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Two Blind Spots in Backtest Validation: From a Manufactured Dollar Index Anomaly to a Corrected Null Result

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Abstract

The standard defence of a quantitative backtest guards against one failure: that too many strategies were tried. Using one instrument and one codebase, we show it misses two others. We first reproduce a sub-period decomposition of the U.S. Dollar Index into overnight and daytime legs, run on a widely used free data source. The result looks compelling: an overnight Sharpe ratio of -1.43, a combined strategy returning 218.8% at Sharpe 1.06, and Newey–West t-statistics significant at $p < 0.001$. Standard overfitting diagnostics do not merely miss the problem, they certify it: the probability of backtest overfitting is 0.02. We then show why. The free source's recorded opening prices carry a stale-print defect, absent from genuine exchange-sourced futures data on the same instrument and sessions, where the overnight leg is statistically indistinguishable from zero (Sharpe $+0.25$, $p = 0.328$) and no strategy in the same grid beats buy-and-hold. A binomial test on the open's location within the session range detects the defect without an external reference ($p = 3.7 \times 10^{-13}$). Separately, a regime-conditioned currency strategy reporting Sharpe 0.72 ($t = 2.78$) reverses once its full 216-configuration search is declared instead of the six originally reported; a correctly specified search of 1,312 configurations finds none exceeding the same search's result on block-bootstrapped returns. Both failures share a structure: standard diagnostics take the data and search space as given, so a stationary defect persists out of sample exactly as a real effect would. We propose a three-part reporting protocol, cheap to apply to any single-instrument backtest, that would have caught both failures at once.

Keywords: backtest overfitting; data quality; deflated Sharpe ratio; foreign exchange; multiple testing; overnight returns; reproducibility; trading strategy evaluation; U.S. Dollar Index

JEL Classification: C12; C15; C52; F31; G14

1. Introduction

The modern defence of a quantitative backtest is a defence against one specific failure. The analyst enumerates the strategies that were tried, then applies a correction for the number of trials: White's Reality Check (White 2000), Hansen's test of superior predictive ability (Hansen 2005), the stepwise procedure of Romano and Wolf (2005), the deflated Sharpe ratio of Bailey and López de Prado (2014), the probability of backtest overfitting of Bailey et al. (2017), or the haircut Sharpe ratio of Harvey and Liu (2015). This machinery is well developed and widely adopted, and a growing empirical literature applies exactly this

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kind of correction to systematic trading strategies, including sub-period decompositions of the type this paper uses (Salotra et al. 2026).

It is also narrow. Every one of these tools takes the input data as given and asks a question about the strategy set. Two failures therefore sit outside its field of view. The input series may be wrong, in which case the correction is applied faithfully to a number that never existed. Or the declared strategy set may be smaller than the set actually explored, in which case the correction is applied to the wrong denominator.

This paper documents one of each, on the same instrument, with one codebase, and shows what a correctly specified analysis of the same market produces once both are fixed. The two failures were not found independently. We first ran a complete overnight/daytime sub-period decomposition of the U.S. Dollar Index (DXY), following the framework Salotra et al. (2026) apply to U.S. sector exchange-traded funds, and obtained what looked like a clean, economically interpretable result: the dollar's entire cumulative return over 2010–2025 accrues during the daytime session, while the overnight session exhibits a large and persistent negative drift. Every diagnostic we knew to run at the time – Newey–West robust t-statistics, mirror-pair reasoning about the effective search size, the deflated Sharpe ratio, White's Reality Check – was applied to that result, and the finding survived some of them. It was only after we tried to source the same series from a second, exchange-based provider, as a routine robustness check, that the overnight effect vanished. We present the full original exercise in Section 3, exactly as it was first run, because the point of this paper is not that vendor data contains errors – Ince and Porter (2006) established that for equity returns two decades ago and their paper remains the standard reference – but what happens when that literature is placed next to the overfitting diagnostics that postdate it. The answer is that those diagnostics invert. On the corrupted series the probability of backtest overfitting is 0.016 to 0.028, which reads as an exceptionally robust result. On the correct series for the same instrument it is 0.765 to 0.779, which reads as pure noise fitting. Both readings are internally correct. A stationary data defect keeps winning out of sample for exactly the same reason a real anomaly does, so no overfitting diagnostic can separate them.

Investigating why the two data sources disagreed also exposed the second failure. A regime-conditioned cross-sectional currency strategy we had set aside as a candidate explanation for part of the overnight pattern reports a Sharpe ratio of 0.72 with a Newey–West t-statistic of 2.78 on six reported configurations. Declaring the full parameter space that was actually available to the search – frequency, cross-sectional slice, cardinality, the regime filter and the momentum lookback – expands that to 216 configurations, and the result does not survive the correction once the true denominator is used.

Section 2 describes the data, including why the U.S. Dollar Index itself is not a tradable instrument and what that implies for which series should be used in an exercise like this one. Section 3 reproduces the naive decomposition in full, on the data source that first motivated this paper. Section 8 shows what the standard diagnostics say about that result, isolates the defect, and gives a screen that detects it using only the series' own prices. Section 9 documents the second failure, an undeclared search space. Section 10 runs a correctly specified search on the validated data and introduces the noise ceiling. Section 11 asks whether anything in this market is profitable once the analysis is driven by structure rather than by enumeration. Section 12 states the reporting protocol we recommend, Section 13 discusses what the episode implies for how economically plausible a spurious result can look, and Section 14 states the limitations.

2. Data

2.1. The Index Is Not Directly Tradable

Before describing the four data sources used in this paper, we state a distinction that turns out to matter for everything that follows. The DXY index is a calculated benchmark: ICE Data Indices, its administrator, computes it from a real-time feed of the six constituent currencies' mid-point spot rates, and states directly that it is not possible to invest in the index itself (ICE Data Indices 2026). There is no security whose price is the index level. Exposure is instead taken through the ICE U.S. Dollar Index futures contract, or through futures-based exchange-traded products such as the Invesco DB US Dollar Index Bullish Fund, which itself holds long positions in the futures contract rather than in the index (Invesco 2026).

This distinction is not merely terminological, and it is the reason this paper exists. A free, widely used redistribution of the DXY *spot* index is easy to obtain and easy to mistake for a tradable series – indeed, an earlier pass of this analysis did exactly that, treating a popular free provider's DXY feed as if it were the futures contract. It is not: as Table 1 shows, the free series and the genuine exchange-sourced futures series are two different products, both nominally tracking the same index, obtained from two different pipelines. The futures price additionally embeds the cost of carry between valuation and settlement (Du et al. 2018), so even a correctly labelled futures series would not equal the spot index level exactly. Section 8 shows that the gap between the free provider's series and the exchange-sourced futures series in this instance is not carry, but a data defect in the free series – and that the mistake of treating the two as interchangeable is precisely what manufactures the anomaly this paper investigates.

2.2. Data Sources

Table 1 lists the four sources. The two U.S. Dollar Index series are the centre of the paper. One is the ICE spot index as redistributed by a free provider, the series on which the original result was found. The other is the ICE Dollar Index futures contract obtained from a commercial vendor sourced from the exchange. They describe the same underlying index over the same dates.

Table 1. Data sources. The two dollar index series in the first two rows describe the same underlying index over the same dates and are the comparison on which Section 8 rests.

Series	Source	Period	Sessions
Dollar index, free provider	Free redistribution of the ICE spot index	2010-12 to 2025-12	3,932
Dollar index futures	Commercial vendor, exchange sourced (DX)	2010-12 to 2025-12	3,889
Currency futures	CME contracts E6, B6, D6, J6, S6	2010-12 to 2025-12	3,800
Spot rates and interest rates	Federal Reserve H.10 and OECD via FRED	1999-01 to 2026-08	6,929

The five currency futures are the CME contracts for the euro, sterling, Canadian dollar, yen and Swiss franc. No liquid Swedish krona contract exists, so the tradeable universe is five of the six index constituents. All five carry median daily volume above 25,000 contracts, contain no violation of the open, high, low, close ordering and are correctly dated: their return correlation with the Federal Reserve noon fix peaks at lag zero.

Nearby continuous futures jump when the front contract changes. That jump is the spread between two contracts and not a return any holder earns, so all returns are computed

within contract and the 61 roll sessions in each series are assigned a return of zero. This also means that contract changes cannot contribute to the overnight leg examined in Section 8.

Every t statistic reported below is corrected for heteroskedasticity and autocorrelation following Newey and West (1987), using a Bartlett kernel with the standard automatic bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$, which gives eight lags at this sample length. The conclusions are insensitive to that choice: the overnight leg of the free series returns $p < 0.0001$ at eight, nine and twelve lags while the exchange series returns 0.328, 0.328 and 0.331.

2.3. Sub-Period Return Definitions

For each day d_i we distinguish two sub-periods. The overnight sub-period runs from the close of day d_{i-1} to the open of day d_i. The daytime sub-period runs from the open of day d_i to the close of day d_i. Following Salotra et al. (2026), we denote the corresponding returns as

$$R_i^{CO} = \frac{O_i}{C_{i-1}} - 1, \tag{1}$$

$$R_i^{OC} = \frac{C_i}{O_i} - 1, \tag{2}$$

where O_i and C_i are the open and close on day d_i. The close-to-close return satisfies

$$1 + R_i^{CC} = (1 + R_i^{CO})(1 + R_i^{OC}), \tag{3}$$

so the decomposition is exhaustive and lossless.

A position of USD 100 is compounded through the sub-period return series. The final balance B_i on day d_i follows from the strategy return R_i and the previous balance as $B_i = (1 + R_i)B_{i-1}$. We report the terminal balance, total return, compound annual growth rate (CAGR), the annualized Sharpe ratio at a zero risk-free rate,

$$\mathrm{SR} = \frac{\bar{R}_{\mathrm{daily}}}{\sigma_{\mathrm{daily}}} \sqrt{252}, \tag{4}$$

annualized volatility, maximum drawdown and the trade count. A trade is counted whenever the target position at the start of a sub-period differs from the position carried into it and is non-zero.

Transaction costs are charged as a proportional deduction of c basis points from the sub-period return on every trade, for $c \in \{0, 1, 2\}$. We stress that $c = 0$ is a diagnostic rather than a scenario: it isolates the statistical signal from the execution problem. Where the underlying instrument is a liquid futures contract – as in the exchange-sourced analysis of Sections 8 onward – $c = 1$ is a reasonable estimate of an all-in institutional round trip, and $c = 2$ is closer to what a retail participant or small fund would face. Section 3 applies the same convention to the free-provider series as a diagnostic benchmark, without implying that series is itself tradable.

2.4. Software and Generative AI Disclosure

All analysis was performed in Python 3 using `pandas`, `NumPy`, `SciPy` and `statsmodels`. Figures were produced with `matplotlib`.

During the preparation of this manuscript, the authors used a large language model assistant for code review of the statistical testing routines and for language editing. The authors independently verified all analytical output, re-ran every reported figure from the raw data, and take full responsibility for the content of this publication.

3. The Naive Decomposition

A growing body of evidence shows that the overnight and daytime portions of a trading day carry different risk premia. In U.S. equities, the overnight session has historically earned the bulk of the market's cumulative return, whereas the daytime session contributed comparatively little (Cliff et al. 2008; Lou et al. 2019). Whether an analogous, or opposite, pattern exists in currency markets motivated the exercise reproduced in this section: a complete sub-period decomposition of the DXY on the free-provider series described in Table 1, run exactly as an analyst without a second, exchange-sourced series to cross-check against would run it. Every number in this section is computed on that series. We present the full apparatus – worked example, complete strategy grid, transaction cost decay, and formal statistical testing – deliberately, because the point of the next section is not that this analysis is careless. It is methodologically ordinary, applies established diagnostics correctly, and still produces a result that does not exist.

3.1. A Worked Example

Before turning to the full sample it is useful to see the mechanics on real data. Table 2 gives the DXY open and close for the week of 27 June 2016, and Table 3 converts those prices into the three returns defined above. The week is a compact illustration of the paper's central fact: compounded across the five sessions the overnight leg loses 0.632%, the daytime leg gains 0.847%, and the 24 h return a passive holder actually receives is the small residual of the two, +0.210%.

Table 2. Example of free-provider DXY data. Daily open O_i and close C_i for the week of 27 June 2016. The close of the preceding session, 24 June 2016, is included because the first overnight return is measured from it.

Date d_i	**Day**	**Open** O_i	**Close** C_i
24 June 2016	Friday	93.45	95.45
27 June 2016	Monday	95.53	96.54
28 June 2016	Tuesday	96.41	96.25
29 June 2016	Wednesday	96.04	95.77
30 June 2016	Thursday	95.67	96.14
1 July 2016	Friday	95.89	95.65

Table 3. The same week decomposed into sub-period returns (%). R_i^{CO} is the overnight (close-to-open) return, R_i^{OC} the daytime (open-to-close) return, and R_i^{CC} the 24 h return, which satisfies $1 + R_i^{CC} = (1 + R_i^{CO})(1 + R_i^{OC})$. Over this week the overnight leg loses 0.63% in total while the daytime leg gains 0.85%, the asymmetry this paper is about.

		Prices		**Returns (%)**		
Date d_i	**Day**	**Open** O_i	**Close** C_i	R_i^{CO} **Night**	R_i^{OC} **Day**	R_i^{CC} **24 h**
27 Jun	Mon	95.53	96.54	0.084	1.057	1.142
28 Jun	Tue	96.41	96.25	−0.135	−0.166	−0.300
29 Jun	Wed	96.04	95.77	−0.218	−0.281	−0.499
30 Jun	Thu	95.67	96.14	−0.104	0.491	0.386
1 Jul	Fri	95.89	95.65	−0.260	−0.250	−0.510
Compounded over the week				−0.632	0.847	0.210

Table 4 shows the position each of five representative strategies holds in each sub-period, and Table 5 compounds USD 100 forward through them. The ordering over this single week already previews the full sample. Strategy 7, which shorts the overnight leg and holds the daytime leg, ends at USD 101.49 against USD 100.21 for buy-and-hold, and

Table 4. Positions held by five key strategies in each sub-period of the example week. The figures beneath each day give that day's realised R_i^{CO} and R_i^{OC} in %. Inertia and reversal rules condition only on information already observable: the overnight position uses the previous daytime return, the daytime position uses the same morning's overnight return.

		Mon		Tue		Wed		Thu		Fri	
#	Strategy	Night 0.08	Day 1.06	Night −0.13	Day −0.17	Night −0.22	Day −0.28	Night −0.10	Day 0.49	Night −0.26	Day −0.25
5	(Long, Long)	long	long	long	long	long	long	long	long	long	long
2	(Short, Cash)	short	cash	short	cash	short	cash	short	cash	short	cash
3	(Cash, Long)	cash	long	cash	long	cash	long	cash	long	cash	long
7	(Short, Long)	short	long	short	long	short	long	short	long	short	long
20	(Short, Reversal)	short	short	short	long	short	long	short	long	short	long

Table 5. Accumulation of USD 100 through the example week, at zero transaction cost. Each day's balance is $B_i = (1 + R_i)B_{i-1}$ with $R_i = (1 + p_i^{CO} R_i^{CO})(1 + p_i^{OC} R_i^{OC}) - 1$, where p is the position from Table 4. Strategy 5 is buy-and-hold in sub-period form.

#	Strategy	Start	Mon	Tue	Wed	Thu	Fri
5	(Long, Long)	100.00	101.14	100.84	100.34	100.72	100.21
2	(Short, Cash)	100.00	99.92	100.05	100.27	100.37	100.63
3	(Cash, Long)	100.00	101.06	100.89	100.61	101.10	100.85
7	(Short, Long)	100.00	100.97	100.94	100.88	101.48	101.49
20	(Short, Reversal)	100.00	98.86	98.83	98.77	99.35	99.36

it does so while being flat on the direction that hurt the passive holder. Strategy 20 ends below 100 because its daytime reversal rule was wrong on the Monday, when a positive overnight return led it to take a short position into a session that rose 1.06%. One week is of course not evidence of anything; it is shown so that every number in the tables that follow can be traced back to a transparent construction.

3.2. Data Limitation

The free-provider DXY series contains 138 days, or 3.5% of the sample, on which the series did not print while the spot currency pairs did. We treat those days as flat for every strategy. The buy-and-hold benchmark is computed on the forward-filled close series so that it reflects the true price ratio $98.24/80.71 = 1.2172$. This works against the strategies, which forgo the missing days' moves. The magnitude is not negligible: computing the benchmark on the same skipped-day convention used for the strategies would place buy-and-hold at USD 104.76 rather than USD 121.72. We use the higher and harder-to-beat figure throughout.

4. Strategies

Each of the two daily sub-periods is assigned one of five elementary rules:

- **Cash**: hold no position;
- **Long**: hold a long position unconditionally;
- **Short**: hold a short position unconditionally;
- **Inertia**: take a long (short) position if the return for the preceding sub-period was non-negative (negative);
- **Reversal**: take a long (short) position if the return for the preceding sub-period was negative (non-negative).

We denote a daily strategy as a pair (overnight, daytime). There are $5 \times 5 = 25$ possible combinations. Excluding the trivial (Cash, Cash) strategy, there are 24 strategies to consider, so that we are always invested in at least one sub-period. All conditioning information is strictly backward looking: the overnight position for day d_i is set from the daytime return

on day d_{i-1}, and the daytime position for day d_i is set from the overnight return on day d_i, which is already observable at the open. The 24 strategies are enumerated in Table 6.

5. Strategy Performance Analysis

5.1. The Dollar Earns Its Return During the Daytime Sub-Period

Figure 1 presents the central empirical fact. Holding DXY only during the open-to-close sub-period grows USD 100 into USD 196.01, a CAGR of 4.41%, a Sharpe ratio of 0.68 and a maximum drawdown of −12.4%. Holding it only overnight shrinks USD 100 to USD 61.02, a CAGR of −3.12% and a Sharpe ratio of −1.43. Buy-and-hold, which is the product of the two legs, ends at USD 121.72.

The overnight leg is notable less for its magnitude than for its consistency. Its annualized volatility is only 2.19%, roughly one third that of the daytime leg, yet its Sharpe ratio of −1.43 is the largest in absolute value anywhere in our set. The mirror strategy, (Short, Cash), delivers a Sharpe ratio of +1.43 with a maximum drawdown of only −5.3%.

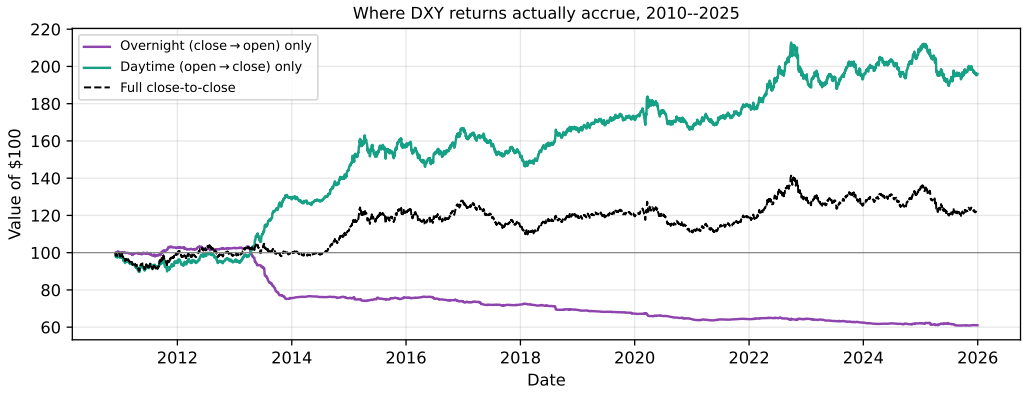


Figure 1. Cumulative value of USD 100 invested in the DXY overnight sub-period only, the daytime sub-period only, and continuously, 2010–2025. The two sub-periods move in opposite directions across the entire sample.

5.2. The Full Strategy Set

Table 6 reports all 24 strategies at zero transaction cost. The best performer is Strategy 7, (Short, Long), which combines the two legs identified above. It ends at USD 318.80, a total return of 218.8%, a CAGR of 7.71%, a Sharpe ratio of 1.06 and a maximum drawdown of −13.0%.

5.3. Mirror-Pair Structure and Effective Degrees of Freedom

The strategy set is not 24 independent experiments. Because each elementary rule has an exact negation within the set, the strategies partition into 12 mirror pairs whose Sharpe ratios are equal or near equal in magnitude and opposite in sign. Pair (1, 2) sits at exactly ∓1.432 and pair (3, 4) at exactly ±0.681. Pair (7, 8) comes out at +1.064 and −1.061, and pair (17, 20) at −0.850 and +0.853. Pairs involving a Cash leg mirror exactly, whereas pairs with two active legs differ in the third decimal because compounding two sub-period returns is not a linear operation.

The maximum of the set therefore does not represent a search over 24 hypotheses. It represents a search over roughly 12 sign choices, and the winner is the one aligned with the two univariate facts established above. This is a meaningful reduction in the effective multiple-testing burden, although Section 7 shows it is not sufficient to rescue the result.

Table 6. All 24 strategies on DXY, 2010–2025, at zero transaction cost. "Final" is the terminal value of USD 100. Strategies are ordered by construction, not by performance.

#	Overnight	Daytime	Final	Total %	CAGR %	Sharpe	MaxDD %	Vol %	Trades
1	Long	Cash	61.02	−39.0	−3.12	−1.432	−41.2	2.19	3932
2	Short	Cash	162.64	62.6	3.17	1.432	−5.3	2.19	3932
3	Cash	Long	196.01	96.0	4.41	0.681	−12.4	6.66	3932
4	Cash	Short	47.60	−52.4	−4.65	−0.681	−60.3	6.66	3932
5	Long	Long	119.61	19.6	1.15	0.203	−14.3	6.80	1
6	Short	Short	77.41	−22.6	−1.63	−0.207	−38.5	6.80	1
7	**Short**	**Long**	**318.80**	**218.8**	**7.71**	**1.064**	**−13.0**	**7.23**	**7864**
8	Long	Short	29.04	−71.0	−7.62	−1.061	−75.8	7.22	7864
9	Cash	Inertia	61.20	−38.8	−3.10	−0.448	−54.9	6.54	3656
10	Cash	Reversal	152.83	52.8	2.76	0.448	−32.0	6.54	3656
11	Inertia	Cash	77.98	−22.0	−1.58	−0.713	−28.4	2.20	3793
12	Reversal	Cash	127.26	27.3	1.56	0.713	−7.9	2.20	3793
13	Inertia	Inertia	47.73	−52.3	−4.63	−0.643	−63.2	6.99	3881
14	Inertia	Reversal	119.18	19.2	1.13	0.199	−38.9	6.81	3706
15	Reversal	Inertia	77.89	−22.1	−1.59	−0.201	−47.2	6.81	3706
16	Reversal	Reversal	194.50	94.5	4.36	0.644	−28.0	7.00	3881
17	Long	Inertia	37.35	−62.7	−6.12	−0.850	−71.3	7.13	3894
18	Long	Reversal	93.26	−6.7	−0.45	−0.034	−44.7	6.66	3971
19	Short	Inertia	99.54	−0.5	−0.03	0.029	−38.6	6.66	3971
20	Short	Reversal	248.56	148.6	6.01	0.853	−29.4	7.14	3894
21	Inertia	Long	152.86	52.9	2.76	0.432	−15.1	6.83	3737
22	Reversal	Long	249.46	149.5	6.03	0.849	−13.7	7.20	4127
23	Inertia	Short	37.12	−62.9	−6.15	−0.846	−69.9	7.20	4127
24	Reversal	Short	60.57	−39.4	−3.16	−0.436	−51.1	6.83	3737
DXY buy-and-hold			121.72	21.7	1.27	0.218	−15.3	6.86	1

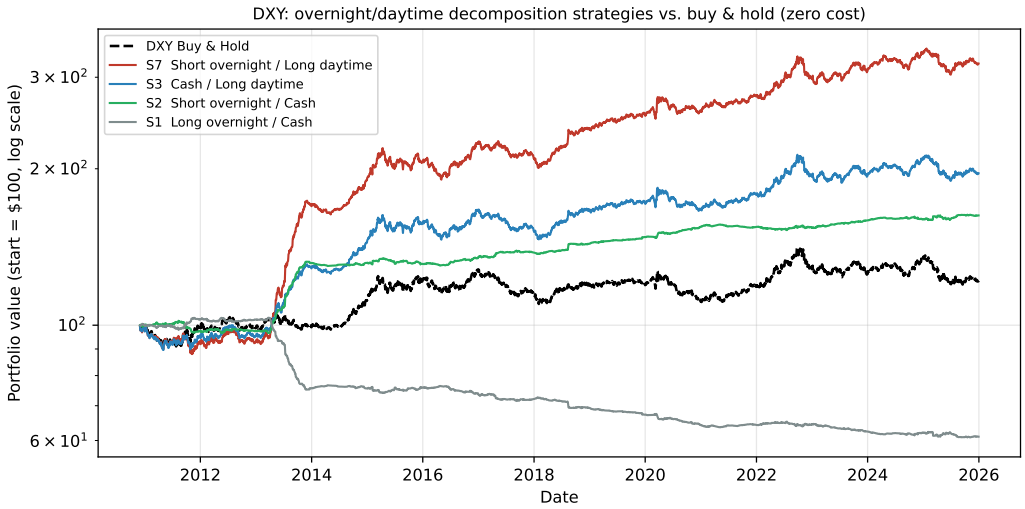


Figure 2. Equity curves for the leading strategies against buy-and-hold, log scale, at zero transaction cost.

5.4. Extension to the Constituent Currency Pairs

Applying the same 24-strategy set to each of the six DXY constituents provides a useful cross-check. The best single pair and strategy combination at zero cost is USD/JPY under Strategy 8, (Long, Short), at USD 170.96 with a Sharpe ratio of 0.42. The sign is inverted relative to the index. Because JPYUSD is quoted in the opposite direction from the index convention, this is the same underlying effect rather than a contradiction, and it indicates that the sub-period asymmetry in DXY is at least partly inherited from the yen leg, which carries the largest overnight interest rate differential in the basket. No constituent pair delivers a gross return within 150 percentage points of the index-level Strategy 7, which suggests that diversification across six pairs is itself contributing to the effect.

6. Transaction Cost Analysis

Strategy 7 changes position twice per day, giving 7864 trades over 3932 sessions. Table 7 and Figure 3 show the consequence.

Table 7. Terminal value of USD 100 for all 24 strategies at three per-trade cost levels. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green marks a strategy that ends above the reference, red one that ends below it, and yellow one that ends within 10% below the reference but on a higher Sharpe ratio.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	61.02	42.33	29.37
2	Short	Cash	162.64	112.84	78.29
3	Cash	Long	196.01	134.13	91.78
4	Cash	Short	47.60	32.57	22.28
5	Long	Long	119.61	119.61	119.61
6	Short	Short	77.41	77.41	77.41
7	Short	Long	318.80	151.36	71.85
8	Long	Short	29.04	13.79	6.54
9	Cash	Inertia	61.20	42.46	29.45
10	Cash	Reversal	152.83	106.03	73.56
11	Inertia	Cash	77.98	54.10	37.53
12	Reversal	Cash	127.26	88.29	61.25
13	Inertia	Inertia	47.73	32.59	22.25
14	Inertia	Reversal	119.18	82.86	57.60
15	Reversal	Inertia	77.89	54.15	37.64
16	Reversal	Reversal	194.50	132.81	90.69
17	Long	Inertia	37.35	25.83	17.87
18	Long	Reversal	93.26	64.00	43.91
19	Short	Inertia	99.54	68.31	46.88
20	Short	Reversal	248.56	171.96	118.96
21	Inertia	Long	152.86	106.48	74.17
22	Reversal	Long	249.46	167.69	112.73
23	Inertia	Short	37.12	24.95	16.77
24	Reversal	Short	60.57	42.19	29.38

Table 7 colours every cell against Strategy 5, the passive (Long, Long) reference, following the convention of Salotra et al. (2026). Nine strategies beat it at zero cost, five at 1 bps and none at 2 bps. The single yellow cell is Strategy 2 at 1 bps: it ends at USD 112.84 against the reference's USD 119.61, so it loses on money, yet its Sharpe ratio of 0.364 is almost double the reference's 0.203. That cell is the entire argument of Section 6 in one place.

1. **The best strategy changes with the cost assumption.** At 0 bps the best strategy is Strategy 7 at USD 318.80. At 1 bps it is Strategy 20, (Short, Reversal), at USD 171.96, because it trades half as often. At 2 bps the best strategy in the entire set is Strategy 5, which is buy-and-hold expressed in sub-period form and trades once. Any claim about the best strategy is therefore conditional on an execution assumption and should be stated as such.
2. **Strategy 7's edge is approximately 1.56 bps per trade.** Its terminal value falls from USD 318.80 to USD 151.36 to USD 71.85 as cost rises from 0 to 1 to 2 basis points. It drops below the buy-and-hold benchmark of USD 121.72 at approximately 1.30 bps and below break-even at approximately 1.56 bps, as shown in Figure 4. At the time, treating this as a liquid tradable futures contract, 1.56 bps looked like a narrow but not obviously unachievable margin – which is why the accurate statement, even setting aside the data defect established in Section 8, is that the naive analysis placed the strategy at the boundary of viability rather than reporting a clean 218.8% return. Section 8 shows the boundary question never actually arises, because the underlying

series is not a tradable futures contract in the first place.

3. **Low volatility is not the same as robustness.** Strategy 2, (Short, Cash), has the highest gross Sharpe ratio in the entire set at 1.432, on the lowest volatility of 2.19% and the shallowest drawdown of −5.3%, and it trades once per day rather than twice. It is tempting to conclude that it is the safer implementation. It is not. Because a fixed per-trade cost is deducted in return space, it consumes a much larger fraction of a low-volatility strategy's return. At 1 bps Strategy 2's Sharpe ratio falls to 0.364, below Strategy 7's 0.404, and at 2 bps it falls to −0.704 against Strategy 7's −0.257. The strategy with the best gross risk-adjusted profile is the one most damaged by frictions. Ranking strategies by gross Sharpe ratio therefore systematically favours precisely those strategies least able to pay for their own execution.

6.1. Risk Metrics Under Cost

Terminal value is only one of the things a cost deduction moves. Tables 8, 9 and 10 repeat the exercise for the annualized Sharpe ratio, annualized volatility and maximum drawdown, coloured against the same passive reference. Reading them together makes the point of observation 3 above visible rather than merely asserted.

Two patterns stand out. First, volatility is almost inert to cost: ten strategies sit below the reference's 6.80% at zero cost and eleven do so at 1 and 2 bps, and the column-to-column changes are in the third decimal. A fixed per-trade deduction shifts the mean of the return series without materially changing its dispersion. Second, Sharpe ratios and drawdowns are not inert at all. Nine strategies beat the reference's Sharpe ratio at zero cost, six at 1 bps and none at 2 bps; the count of strategies with a shallower drawdown falls from five to one to zero over the same range. The strategies that lose their green first are, in every case, the ones that trade twice a day.

The clearest single case is Strategy 2, (Short, Cash). At zero cost it has the highest Sharpe ratio in the set at 1.432, the lowest volatility at 2.19% and the shallowest drawdown at −5.3%. It is the safest-looking strategy in the table on every risk measure at once. By 2 bps its Sharpe ratio is −0.704, worse than Strategy 7's −0.257, and its drawdown has deepened from −5.3% to −31.8%. Its volatility, meanwhile, has not moved: 2.19% at every cost level. The risk did not increase. The return available to pay for the risk disappeared, and a low-volatility strategy has proportionally less of it to give away.

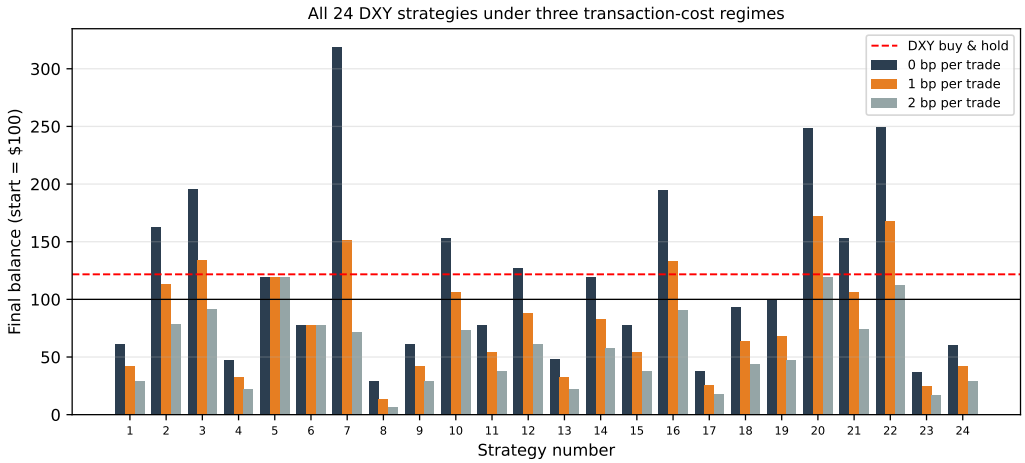


Figure 3. Terminal value of USD 100 for all 24 strategies at 0, 1 and 2 basis points per trade. The dashed line is DXY buy-and-hold.

Table 8. Annualized Sharpe ratio for all 24 strategies at three per-trade cost levels. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green marks a higher Sharpe ratio than the reference, red a lower one.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	−1.432	−2.498	−3.561
2	Short	Cash	1.432	0.364	−0.704
3	Cash	Long	0.681	0.316	−0.049
4	Cash	Short	−0.681	−1.045	−1.410
5	Long	Long	0.203	0.203	0.203
6	Short	Short	−0.207	−0.207	−0.207
7	Short	Long	1.064	0.404	−0.257
8	Long	Short	−1.061	−1.722	−2.384
9	Cash	Inertia	−0.448	−0.806	−1.164
10	Cash	Reversal	0.448	0.090	−0.268
11	Inertia	Cash	−0.713	−1.777	−2.839
12	Reversal	Cash	0.713	−0.352	−1.416
13	Inertia	Inertia	−0.643	−0.990	−1.335
14	Inertia	Reversal	0.199	−0.143	−0.487
15	Reversal	Inertia	−0.201	−0.542	−0.881
16	Reversal	Reversal	0.644	0.295	−0.055
17	Long	Inertia	−0.850	−1.178	−1.504
18	Long	Reversal	−0.034	−0.397	−0.761
19	Short	Inertia	0.029	−0.333	−0.692
20	Short	Reversal	0.853	0.523	0.192
21	Inertia	Long	0.432	0.093	−0.246
22	Reversal	Long	0.849	0.496	0.143
23	Inertia	Short	−0.846	−1.199	−1.550
24	Reversal	Short	−0.436	−0.775	−1.112

Table 9. Annualized volatility (%) for all 24 strategies at three per-trade cost levels. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green marks lower volatility than the reference, red higher volatility.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	2.19	2.20	2.20
2	Short	Cash	2.19	2.19	2.19
3	Cash	Long	6.66	6.66	6.66
4	Cash	Short	6.66	6.66	6.66
5	Long	Long	6.80	6.80	6.80
6	Short	Short	6.80	6.80	6.80
7	Short	Long	7.23	7.23	7.23
8	Long	Short	7.22	7.22	7.22
9	Cash	Inertia	6.54	6.54	6.54
10	Cash	Reversal	6.54	6.54	6.54
11	Inertia	Cash	2.20	2.20	2.20
12	Reversal	Cash	2.20	2.20	2.20
13	Inertia	Inertia	6.99	7.01	7.03
14	Inertia	Reversal	6.81	6.80	6.79
15	Reversal	Inertia	6.81	6.82	6.84
16	Reversal	Reversal	7.00	6.99	6.97
17	Long	Inertia	7.13	7.15	7.17
18	Long	Reversal	6.66	6.65	6.64
19	Short	Inertia	6.66	6.68	6.69
20	Short	Reversal	7.14	7.12	7.11
21	Inertia	Long	6.83	6.83	6.83
22	Reversal	Long	7.20	7.20	7.20
23	Inertia	Short	7.20	7.20	7.21
24	Reversal	Short	6.83	6.84	6.85

Table 10. Maximum drawdown (%) for all 24 strategies at three per-trade cost levels. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green marks a shallower drawdown than the reference, red a deeper one.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	−41.2	−58.0	−70.8
2	Short	Cash	−5.3	−9.1	−31.8
3	Cash	Long	−12.4	−16.9	−30.5
4	Cash	Short	−60.3	−71.9	−80.1
5	Long	Long	−14.3	−14.3	−14.3
6	Short	Short	−38.5	−38.5	−38.5
7	Short	Long	−13.0	−20.2	−49.8
8	Long	Short	−75.8	−87.5	−93.9
9	Cash	Inertia	−54.9	−64.7	−72.4
10	Cash	Reversal	−32.0	−46.5	−58.2
11	Inertia	Cash	−28.4	−48.1	−63.2
12	Reversal	Cash	−7.9	−16.7	−38.8
13	Inertia	Inertia	−63.2	−71.3	−77.9
14	Inertia	Reversal	−38.9	−52.5	−63.1
15	Reversal	Inertia	−47.2	−57.4	−66.5
16	Reversal	Reversal	−28.0	−40.5	−53.6
17	Long	Inertia	−71.3	−77.6	−82.6
18	Long	Reversal	−44.7	−56.9	−66.4
19	Short	Inertia	−38.6	−46.8	−57.7
20	Short	Reversal	−29.4	−36.4	−49.1
21	Inertia	Long	−15.1	−20.6	−36.7
22	Reversal	Long	−13.7	−16.1	−24.5
23	Inertia	Short	−69.9	−78.9	−85.2
24	Reversal	Short	−51.1	−63.6	−73.7

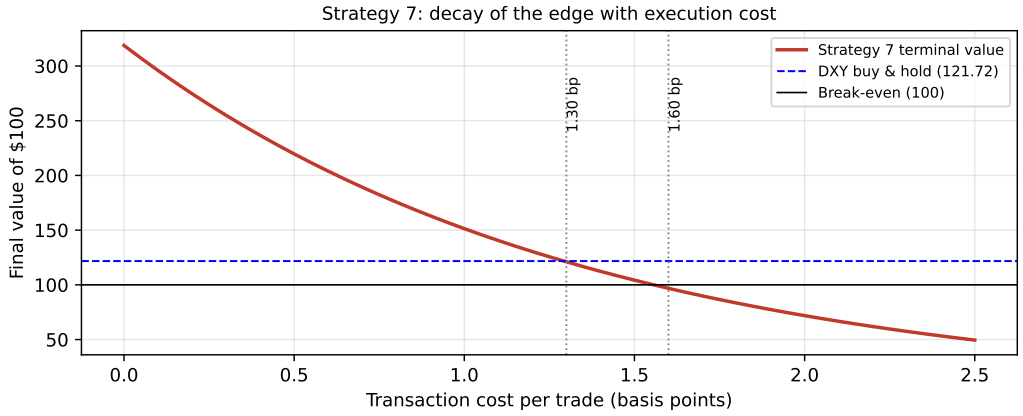


Figure 4. Terminal value of Strategy 7 as a continuous function of per-trade cost. The edge falls below buy-and-hold at 1.30 bps and below break-even at 1.56 bps.

7. Robustness and Statistical Validation

A terminal balance is not evidence. This section asks whether the results above survive formal testing. In one important respect they do not.

7.1. Are the 24 Strategies Distinguishable?

A one-way ANOVA on mean daily return across the 24 strategies rejects equality of means, with $F(23, 94344) = 6.97$ and $p < 0.0001$. However, every assumption behind that test fails on this data. Levene's test rejects equal variances at $p < 10^{-16}$, Jarque–Bera rejects normality of the winning strategy's returns with $JB = 2341.7$ at $p < 10^{-16}$, and Ljung–Box rejects independence at ten lags with $p < 0.0001$. We therefore also report the distribution-free Kruskal–Wallis analogue, which agrees: $H = 113.15$ at $p < 0.0001$.

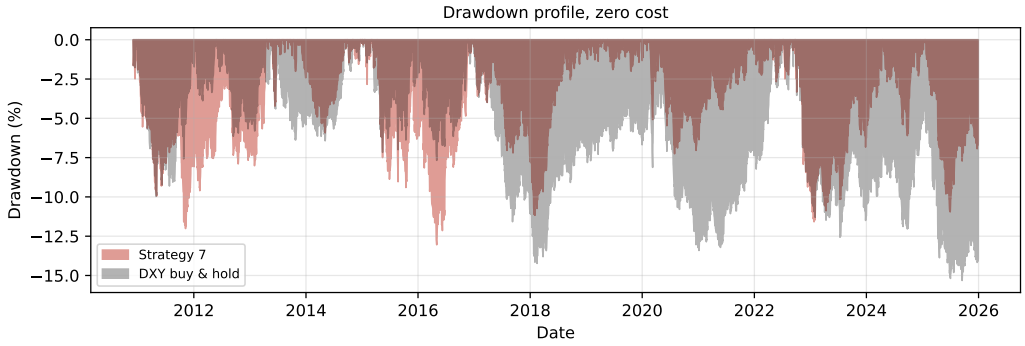


Figure 5. Drawdown profile of Strategy 7 against buy-and-hold at zero cost. Strategy 7's maximum drawdown of −13.0% is slightly shallower than the benchmark's −15.3% despite a 7.71% CAGR.

We include the ANOVA for completeness and to state explicitly that it is the wrong instrument for this question. It establishes only that the strategies differ from one another, which is guaranteed by construction, since twelve of them are the exact negations of the other twelve.

7.2. Is the Best Strategy's Mean Return Significantly Positive?

Table 11 reports Newey–West heteroskedasticity and autocorrelation consistent t-statistics on mean daily returns, using nine lags and a Bartlett kernel (Newey and West 1987), together with the Probabilistic Sharpe Ratio of Bailey and López de Prado (2012) and confidence intervals from a stationary block bootstrap using 5000 resamples and 21-day blocks.

Table 11. Formal tests on the leading strategies. t_{NW} is the Newey–West t-statistic on mean daily return. PSR_b is the probability that the true Sharpe ratio exceeds the benchmark's 0.218. DSR is the Deflated Sharpe Ratio, correcting for 24 trials, skewness, kurtosis and sample length.

#	Strategy	SR	t_{NW}	p	PSR_b	DSR
7	(Short, Long)	1.064	3.681	0.0002	0.9996	**0.046**
2	(Short, Cash)	1.432	4.317	1.6×10^{-5}	1.0000	0.405
22	(Reversal, Long)	0.849	3.104	0.0019	0.9940	0.006
20	(Short, Reversal)	0.853	2.906	0.0037	0.9946	0.006
3	(Cash, Long)	0.681	2.601	0.0093	n/a	n/a
n/a	DXY buy-and-hold	0.218	0.891	0.373	n/a	n/a

Read in isolation these are strong results. Strategy 7's mean daily return is significant at $p = 0.0002$ after correcting for autocorrelation and heteroskedasticity, and its bootstrapped 95% confidence interval for the annualized Sharpe ratio is $[0.490, 1.699]$, comfortably excluding zero. The buy-and-hold benchmark, by contrast, is not statistically distinguishable from zero at all, with $t = 0.891$ and $p = 0.373$. Strategy 2's interval is $[0.774, 2.061]$.

7.3. Correcting for the Search

The tests above condition on having chosen the strategy in advance. We did not. We selected the maximum of a 24-strategy set, and the appropriate correction is the Deflated Sharpe Ratio of Bailey and López de Prado (2014), which asks whether an observed Sharpe ratio exceeds what the maximum of N trials would be expected to produce under a null of zero true skill, given the observed dispersion of Sharpe ratios across those trials.

The dispersion across our set is very large. Sharpe ratios range from −1.432 to +1.432, which pushes the expected maximum under the null to 1.486 in annualized terms. That is higher than Strategy 7's observed 1.064. The resulting Deflated Sharpe Ratio is 0.046, below the conventional 0.05 threshold. We therefore cannot reject the hypothesis that Strategy 7's

performance is what selecting the best of 24 dispersed trials would produce by chance alone. White's Reality Check (White 2000) delivers the same verdict more directly at $p = 0.062$.

7.4. Interpretation

Two considerations cut against the pessimistic reading and one reinforces it.

The Deflated Sharpe Ratio assumes N independent trials. Our set is not independent. It consists of 12 mirror pairs, so the distribution of Sharpe ratios is symmetric about zero by construction. That symmetry roughly doubles the variance term and therefore inflates the null's expected maximum. A defensible alternative is to treat the effective number of trials as 12. Furthermore, the t-statistics in Table 11 are computed on the univariate legs identified in Section 5 rather than on a pattern discovered by scanning. Strategy 7 was not found by search; it is the arithmetic composition of two facts that are individually significant.

Against this, Strategy 2 has a Deflated Sharpe Ratio of 0.405, far above Strategy 7's, precisely because its Sharpe ratio of 1.432 sits closer to the null's expected maximum. If one insists on a single result from this study that clears formal multiple-testing correction, it is the overnight leg alone rather than the headline combination.

At this point in the analysis, run on the free-provider series alone, the weakest defensible conclusion appeared to be that the sub-period asymmetry in DXY was statistically robust even if the specific 218.8% figure attached to Strategy 7 was not. That conclusion turns out to be wrong, and Section 8 shows why: it is not that the asymmetry survives on weaker grounds than the headline number, but that the asymmetry itself is not present in genuine exchange-sourced data on the same instrument over the same sessions. Every diagnostic in this section is doing exactly what it is designed to do; none of them is designed to ask whether the overnight leg's Sharpe ratio of -1.432 reflects real prices.

8. A Data Defect That the Diagnostics Certify

8.1. The result and the first attempt to explain it

Decomposing the daily return into an overnight leg from the previous close to the open and a daytime leg from the open to the close produces, on the free series, an overnight leg with a Sharpe ratio of -1.36 and $p < 0.0001$. This is a large effect, and it is essentially the same effect reported as Strategy 1's overnight leg in Section 3 (Sharpe -1.432 on the full 3,932-session sample; the small difference in magnitude is sample overlap, since the comparison here is restricted to the 3,793 sessions common to both series). On the exchange futures for the same instrument the overnight leg has a Sharpe ratio of $+0.25$ with $p = 0.328$. Both figures are computed on the 3,793 overnight returns available in the 3,794 sessions for which both series report a complete bar, so the two are directly comparable.

A naive way to locate the source is to substitute one vendor's opening prices into the other vendor's series. That test is invalid here. The two series carry a persistent level offset of 0.041 index points, which is 6.7 per cent of a typical daily range, so a price level substitution injects the offset into every return. Performing it produced a terminal value of 1178 for one strategy, which is not a finding but an artefact of the offset.

8.2. Isolating the open

The valid version transfers the close to open *return* rather than the price, so that each row uses one vendor's price level throughout and the result is free of the offset. Table 12 reports it.

Table 12. Scale free decomposition. Each row uses a single vendor's price level throughout and transfers only the close to open return, so the level offset between vendors cannot contribute. The daytime leg is the residual, which makes the 24 hour return identical across rows by construction. Terminal values start from 100 and the sample is the 3,793 overnight returns common to both series.

	Overnight leg			Daytime leg		
Price level, opening return	**Final**	**Sharpe**	p	**Final**	**Sharpe**	p
Free, free open	61.68	−1.358	0.0000	197.35	0.700	0.0086
Free, exchange open	108.08	+0.252	0.3280	112.62	0.151	0.5534
Exchange, exchange open	108.08	+0.252	0.3280	112.22	0.146	0.5668
Exchange, free open	61.68	−1.358	0.0000	196.65	0.678	0.0103

The overnight leg follows whichever vendor's open is used and is unchanged by whose closes are used. This is as close to a controlled experiment as observational data allows. The opening prices are the entire source of the effect.

Table 13 shows what is wrong with them. With the level offset removed, the free series' open lies outside the exchange's true high to low range on 7.30 per cent of sessions. On the 204 sessions where the free open equals its own daily low it sits below the exchange low, at −0.073 of the exchange range and falls outside that range on 56.9 per cent of those sessions against 3.5 per cent on ordinary sessions. A price outside the session range is not a price at which the instrument traded. The free series also records an open exactly equal to the previous close on 6.84 per cent of sessions against 1.34 per cent for the exchange series, which is the signature of a stale print.

Table 13. Location of the recorded open within the exchange session range, with the level offset removed. A value of 0 is the exchange low and 1 the exchange high. The sessions on which the free open equals its own extreme are the sessions that carry the anomaly.

Session group	n	**Open location**	**Outside range**
Free open at its own low	204	−0.073	56.9%
Free open at its own high	82	+0.994	45.1%
All other sessions	3,508	+0.494	3.5%
Exchange open, all sessions	3,889	+0.498	0.00%

8.3. What the diagnostics say

The 24-strategy grid of Section 3 (Table 6) was rerun on both series. On exchange data no strategy beats buy and hold at any cost level, the analysis of variance across the 24 gives $F(23, 93312) = 0.306$ with $p = 0.9994$ and the largest Sharpe ratio in the grid, +0.269, is below the +0.316 that the null expects the maximum of 24 trials to produce. The walk forward over 13 out of sample years ends at 97.69 against buy and hold at 122.64. There is nothing in the exchange data.

Table 14 is the central result of this section. Applied to the corrupted series at zero cost, Hansen's test rejects at $p = 0.013$, three configurations survive the stepwise procedure of Romano and Wolf (2005) and the probability of backtest overfitting is 0.016. Applied to the correct series the same diagnostics report 1.000, zero survivors and 0.779. The diagnostics rank the false result far above the true one.

Table 14. Multiple testing and overfitting diagnostics applied to the same 24 strategy grid on the two dollar index series, at zero transaction cost. RW is the number of configurations surviving the stepwise familywise error procedure of Romano and Wolf (2005). PBO is the probability of backtest overfitting from the combinatorially symmetric cross validation of Bailey et al. (2017), reported at 12 and 16 blocks. Lower PBO indicates a result that persists out of sample.

Series	White RC p	Hansen SPA p	RW survivors	PBO
Free provider (the artefact)	0.1032	0.0132	3 of 24	0.016 to 0.028
Exchange futures (correct)	0.9924	1.0000	0 of 24	0.765 to 0.779

Both readings are correct on their own terms. The probability of backtest overfitting asks whether a strategy selected in sample continues to win out of sample. A persistent data defect continues to win, because it is present in both halves of the sample. To every overfitting diagnostic a stationary artefact and a real anomaly are the same object. This is a structural limitation of the entire family of tools and not a defect in any one of them.

8.4. A screen that requires no reference data

The comparison above required a second, trustworthy series. Most analysts do not have one. We therefore propose a test that uses only the series being examined.

Under the null that a recorded open is an unbiased draw from the prices visited during the session, the open is equally likely to coincide with the session low as with the session high. Counting the two events gives an exact binomial test with no free parameters:

$$n_{\text{low}} \sim \text{Binomial}\left(n_{\text{low}} + n_{\text{high}}, \tfrac{1}{2}\right). \tag{5}$$

Table 15 applies Equation (5) to eight series. It clears the exchange series at $p = 0.788$ and flags the corrupted one at $p = 3.7 \times 10^{-13}$.

Table 15. The open location binomial screen of Equation (5) applied to eight series. Mean position is the average location of the open within the session range, where 0 is the low and 1 the high. The screen clears the series known to be sound and rejects the series known to be corrupt, using only that series' own open, high and low.

Series	At low	At high	Binomial p	Mean position
Dollar index futures, exchange	64	60	0.788	0.499
Dollar index, free provider	204	82	3.68×10^{-13}	0.479
EURUSD, free provider	90	56	0.0061	0.501
GBPUSD, free provider	69	58	0.375	0.504
JPYUSD, free provider	125	103	0.164	0.489
CADUSD, free provider	66	52	0.231	0.506
CHFUSD, free provider	62	44	0.098	0.505
SEKUSD, free provider	44	45	1.000	0.508

Two honest qualifications. The screen also flags EURUSD at $p = 0.0061$, which barely survives a Bonferroni correction across the eight tests at a threshold of 0.00625. We independently established that these spot series are date stamped one trading day late, which is consistent with a true positive, but we cannot prove it is not a multiplicity artefact. And the screen is validated here against exactly one instrument with a known good reference. It is a promising diagnostic and not an established one.

We note also that the mean position of the open within the range, at 0.479 for the corrupted series against 0.499 for the exchange series, is a weak discriminator. The count of extremes carries the information, not the average location.

9. An Undeclared Search Space

The second failure needs no bad data. A regime conditioned cross sectional currency strategy, in which six currencies are ranked each quarter and the two weakest are held with a sign set by the 126 day momentum of the dollar index, reports a Sharpe ratio of 0.72 with a Newey–West t statistic of 2.78 and $p = 0.0055$ on Federal Reserve spot data.

Three defects in the committed implementation had to be corrected first. The pandas call `to_period('2Q')` silently returns quarterly periods, so the reported semiannual row was a byte identical duplicate of the quarterly row. The grid loop was wrapped in a bare exception handler, which concealed that the annual frequency alias had been renamed between pandas versions, so every annual configuration was silently discarded. The construction `np.where(mom < 0, 1, -1)` maps a missing value to -1, so the first 126 sessions, in which no momentum can yet be computed, were traded as though the dollar were rising; correcting this alone moves the headline Sharpe ratio from 0.54 to 0.46. And the constituent price files are date stamped one trading day late relative to the index file.

Table 16 reports what happens when the search space is declared in full. The original account reported six configurations. Frequency, cross sectional slice, cardinality, the regime filter and the momentum lookback are all free parameters and enumerating them gives 144 configurations on the tradeable five currency panel and 216 on the six currency panels.

Table 16. Multiple testing over the declared search space. The strategy is a self financing long and short book, so the correct null is zero excess return; the committed code tested against a long dollar position, which is shown for comparison. The number after each universe is the count of currencies in it. No configuration survives the stepwise procedure in any universe once the full search is declared.

			Null: zero		**Null: long dollar**		
Universe	**Configs**	**Sharpe**	**RC** p	**SPA** p	**RC** p	**SPA** p	**DSR**
Exchange futures, 5	144	0.458	0.2333	0.3833	0.6730	0.7647	0.227
Futures plus krona, 6	216	0.498	0.2203	0.2830	0.4373	0.5933	0.217
Federal Reserve spot, 6	216	0.722	0.0963	0.1570	0.3173	0.4727	0.612

The deflated Sharpe ratio is a confidence level and must exceed 0.95. It reaches 0.612 at best. The expected maximum Sharpe ratio under the null across 144 configurations is $+0.654$, above the $+0.458$ actually observed on tradeable instruments. With the regime correctly left undefined over its own lookback window, no universe clears a conventional threshold on any test.

The configuration is also a spike rather than a plateau. Holding everything else at the reported values on the tradeable five currency panel and moving one parameter at a time, the regime lookback gives -0.319 with no overlay, $+0.136$ at 63 days, $+0.458$ at 126 days and $+0.259$ at 252 days. The rebalance frequency gives -0.234 daily, -0.081 weekly, $+0.047$ monthly, $+0.458$ quarterly, $+0.112$ semiannually and $+0.456$ annually. Only two of six frequencies are materially positive and both neighbours of the reported lookback lose most of the effect, which describes a noisy surface rather than a mechanism.

Finally, an ensemble that holds every configuration in a family at equal weight selects nothing and therefore requires no correction. Averaging the net returns of the 36 regime signed configurations in the bottom slice, which charges each member its own full turnover and so is conservative, gives a Sharpe ratio of 0.148 with $p = 0.57$. Widening the ensemble to all 108 regime signed configurations gives -0.160 with $p = 0.53$.

10. A Correctly Specified Search

10.1. Validating the harness before believing it

Before running any search we validated the evaluation code itself, because a paper about invalid backtests should demonstrate that its own backtest is valid. Table 17 lists the ten checks. The most important is the first. Weights formed from the same day's realised return, applied without the one period shift, must produce an absurd Sharpe ratio; if they do not, the shift is broken and nothing downstream means anything.

Table 17. Validation of the evaluation harness. All ten checks pass; the eight rows below group the four turnover and statistic checks into two lines. The lookahead injection is the critical one, because the harness must be demonstrably capable of detecting a strategy that cheats.

Check	Result
Perfect foresight weights applied without the shift	Sharpe +24.34
The same weights applied through the harness	Sharpe −0.11
One extra lag applied to 14 real families	none collapses
Full search on simulated zero drift returns, RC p	0.20, 0.07, 0.69, 0.38
Observed best of N on noise against theory	0.823 against 0.846
Sharpe, Newey–West t, PSR, cost arithmetic	reproduced to machine precision
Buy and hold turnover; daily sign flipper turnover	0.00; exactly 2.0 per session
Winning configuration rebuilt from first principles	maximum difference 0.0e+00

10.2. The search and the noise ceiling

We evaluated 1,312 configurations built from published specifications: time series momentum (Moskowitz et al. 2012), cross sectional currency momentum (Menkhoff et al. 2012), carry (Koijen et al. 2018; Lustig et al. 2011), value (Asness et al. 2013), moving average rules (Brock et al. 1992), channel breakouts, open interest growth (Hong and Yogo 2012) and volatility scaling (Moreira and Muir 2017), each crossed with four rebalance frequencies, the regime overlay at three lookbacks and with or without volatility targeting. Costs are charged at one basis point per unit of turnover.

The natural benchmark is not zero. It is the Sharpe ratio this same search attains on returns that contain no predictable structure. We construct that benchmark by resampling the actual returns in blocks using the stationary bootstrap of Politis and Romano (1994), which preserves the marginal distributions, the fat tails and the cross sectional correlation while destroying the time series structure a signal could exploit, then running the entire search on the resampled panel. We call the resulting maximum the *noise ceiling*. It is not a new statistic. It is a Monte Carlo reality check expressed in the units a practitioner reads.

Table 18. The correctly specified search against its noise ceiling. The ceiling is the best Sharpe ratio the identical 1,312 configuration search attains on block bootstrapped returns, averaged over six resamples. No configuration on the real data exceeds it and every correction agrees.

Quantity	Value
Best configuration on the real data, net of costs	+0.653
Noise ceiling, Gaussian null	+0.752
Noise ceiling, block bootstrap of the real returns	+0.712
Expected best of 1,312 under the null	+0.865
Configurations exceeding the ceiling	**0 of 1,312**
White Reality Check p	0.659
Hansen SPA p	0.773
Romano–Wolf survivors	0
Deflated Sharpe ratio (needs > 0.95)	0.203
Harvey–Liu adjusted p, Bonferroni and BHY	1.000, 1.000
Probability of backtest overfitting	0.556

Table 18 gives the result. The best configuration found in fifteen years of exchange data is *below* what the same search finds in random data. The holdout is consistent: selecting on 2011 to 2018 and evaluating once on 2019 to 2025, the in sample winner ranks 163rd of 1,312 out of sample and the top ten in sample configurations average a Sharpe ratio of +0.019.

Family ensembles, which select nothing, are reported in Table 19. Not one is significant in either direction.

Table 19. Family ensembles, equal weight within each family, costed in weight space so that trades offsetting across members are not charged. Selecting nothing means no correction is required, so these are the fair test of whether a family of ideas has value.

Family	Sharpe	p	Family	Sharpe	p
Cross sectional momentum	+0.218	0.37	Time series momentum	−0.214	0.37
Value	+0.204	0.43	Volume	−0.232	0.37
Carry	+0.135	0.59	Breakout	−0.289	0.23
Reversal	+0.133	0.58	Moving average	−0.341	0.16
Open interest	+0.094	0.72	Cross sectional reversal	−0.360	0.14
All 1,312 configurations				−0.176	0.48

11. Is Anything Here Profitable?

An exhaustive search that finds nothing invites the objection that the search was the wrong instrument. We therefore also proceeded analytically. The structure of the data is unambiguous. The first order autocorrelation of returns is between −0.004 and +0.033, variance ratios lie between 0.83 and 1.02 and an autoregression of returns on their own lag gives R^2 between 0.0000 and 0.0011. The same autoregression on log realised volatility gives R^2 between 0.966 and 0.972. Volatility is roughly three orders of magnitude more forecastable than direction.

If direction cannot be forecast then only a genuine risk premium or a relationship true by construction can pay. We tested both. The dollar index is an exact deterministic function of its six constituents and the index future shares an expiry code with the five currency futures on 98.2 per cent of sessions, so a matched synthetic basket is a no arbitrage relationship. It is also efficiently priced: the convergence trade returns a Sharpe ratio of +0.09 gross and −0.20 at one basis point per leg, because six legs are required against a spread whose daily change has a standard deviation of five basis points.

Carry is the more serious candidate. The roll gap between the expiring and deferred contract is the market's own quoted interest differential, which we validated against Federal Reserve interbank differentials to within 0.43 percentage points per currency. On the fifteen year futures panel it returns a Sharpe ratio of 0.377 and the mechanism verifies: of the 1.34 per cent annual return, 1.10 per cent is the differential actually held and 0.23 per cent is spot movement, which is the forward premium puzzle behaving as described. Table 20 shows why we do not claim it.

Table 20. Term structure carry on two samples. The futures panel begins in December 2010, immediately after the 2008 carry drawdown. Extending the sample to April 2002 using Federal Reserve spot and OECD interbank rates for all six constituents reverses the sign.

Sample	Currencies	Sharpe
Futures panel, 2011 to 2025	5	+0.377
Federal Reserve spot, 2002 to 2026	6	−0.249
Sub periods of the long sample		
2002 to 2007	6	+0.418
2008 to 2013	6	−0.760
2014 to 2019	6	−0.246
2020 to 2026	6	−0.022

The canonical factor set on the long sample gives carry −0.249, momentum +0.028 and value +0.096, with an equal risk combination of −0.085, against a noise ceiling whose ninetieth percentile is +0.41. Volatility targeting, the one quantity that is forecastable, changes the Sharpe ratio by −0.009 on average across seven series over 27 years and improves three of them, which is inside the bootstrap null. What it does deliver is a reduction in the volatility of realised volatility from 0.025 to 0.015. That is risk control and not return generation.

We report no profitable strategy. The reason is structural rather than a failure of effort. There is no directional predictability, the only candidate risk premium in this universe is negative over the full available sample and five to six highly correlated currencies with a common factor explaining 54.8 per cent of variance cannot support a cross sectional strategy.

12. A Reporting Protocol

Three additions, none expensive.

Screen the inputs. Report the binomial statistic of Equation (5), the frequency with which the open equals the previous close and the count of sessions violating the open, high, low, close ordering. These cost one pass over the data and would have stopped the first failure before any strategy was run.

Declare the whole search. Report every parameter that was varied and the total count of configurations evaluated, including those discarded. Report the ensemble across each family alongside the selected configuration, since an ensemble selects nothing and needs no correction.

Report the noise ceiling. Run the complete search once on block bootstrapped returns and report the best Sharpe ratio it produces. Any result below that number is indistinguishable from luck at that search size. In this study the ceiling was 0.71 to 0.75 for the 1,312 configuration search, 0.38 for a five strategy carry family and 0.41 for the canonical factor set. Every real result fell below its own ceiling.

13. Discussion: A Plausible Story Is Not Evidence

Before the defect in Section 8 was identified, we had a ready set of economic explanations for the overnight leg's negative drift. It is worth recording them, because the fact that they were easy to construct is itself the paper's point.

Carry. The DXY basket is dominated by EUR at 57.6% and JPY at 13.6%. Over most of 2010–2025, U.S. policy rates exceeded euro area and Japanese rates, so a long dollar position accrued positive carry, and a mechanism by which that carry is realised disproportionately outside the daytime session is not implausible on its face. This is consistent with the currency risk premium literature (Du et al. 2018; Lustig et al. 2011), which identifies the interest rate differential as the dominant source of expected currency returns.

Liquidity provision. The close-to-open interval spans the thinnest liquidity window for USD-denominated risk. If dealers demand compensation for holding inventory through it, the price at the open should be systematically displaced relative to the prior close in the direction that penalizes the inventory holder.

Settlement mechanics. A contract with a defined settlement window can exhibit return patterns near that window that have nothing to do with an economic risk premium.

Each of these stories is independently reasonable, each is grounded in real literature, and none of them is why the overnight leg in Section 3 has a Sharpe ratio of -1.432. The actual explanation, established in Section 8, is that a meaningful fraction of the free provider's recorded opening prices are stale prints that fall outside the session's true trading range. No carry mechanism, liquidity story, or settlement-window argument is needed, because there is no effect to explain: the same leg on genuine exchange-sourced data is statistically indistinguishable from zero.

We flag this not to criticise the exercise of proposing mechanisms – it is a normal and useful part of interpreting a backtest – but because the ease with which one can be proposed is exactly what makes a data defect dangerous. A result that survives an economically literate narrative feels more credible than one that does not, and that feeling is uncorrelated with whether the underlying prices are real. The two failures documented in this paper – uninspected inputs and undeclared search spaces – both produce results that are easy to rationalise after the fact. The discipline that catches them has to come from the diagnostics in Sections 8 and 10, not from the plausibility of the story attached to the result.

14. Limitations

We flag five, roughly in order of scope.

The evidence is one currency complex and two vendors. Nothing here generalises to free data as a category or to this vendor across other instruments. The mechanism behind the defect is inferred rather than observed: a stale print, a first electronic tick and a misfielded session extreme are all consistent with what we see and we cannot inspect the vendor's pipeline. The commercial series is itself a redistributor rather than the exchange and its closing field is a last price whose settlement status we could not confirm. The screen of Section 8 is validated against a single instrument, and the noise ceiling of Section 10 is a presentation of an existing idea rather than a new statistic. Finally, the absence of a profitable strategy in Section 11 is a statement about six daily G10 series over fifteen to twenty-seven years and not about foreign exchange markets in general; emerging market carry, options and intraday data all lie outside what we can test.

Two further limitations concern the naive decomposition in Section 3, and remain worth stating even though that section's headline result does not survive Section 8. Costs there are modeled rather than measured: we deduct a flat per-trade proportional cost and do not model bid-ask spread variation, slippage at the open, or market impact. Execution at the open and close is assumed throughout, at prices that a systematic participant cannot

generally guarantee. Neither limitation changes the paper's conclusion, since the naive result fails for a more basic reason before either one becomes relevant, but both would matter for any future backtest that clears the data and search-space screens proposed in Section 12 and needs to be evaluated on its economics.

15. Conclusions

Two failures sit outside the reach of the standard defence of a backtest, and this paper found both on the same instrument while investigating what first looked like a clean empirical result. A sub-period decomposition of the DXY, run on a widely used free data source and validated with the field's standard toolkit – Newey–West robust inference, mirror-pair reasoning, the deflated Sharpe ratio, White's Reality Check – produced a strategy returning 218.8% with a Sharpe ratio of 1.06 and a probability of backtest overfitting of 0.02, a result that by every established measure looked exceptionally robust. It was not. The same instrument on genuine exchange-sourced futures data shows no overnight effect (Sharpe $+0.25$, $p = 0.328$) and no strategy in the same 24-strategy grid beats buy-and-hold. A corrupted input is not merely invisible to the overfitting diagnostics, it is actively certified by them, because a stationary data defect persists out of sample exactly as a real anomaly does. A one-line binomial test on the location of the recorded open within its own session range would have caught it before any strategy was run.

Investigating that discrepancy surfaced a second, unrelated failure. A regime-conditioned currency strategy reporting $t = 3.09$ on six reported configurations fails every standard test once its true 216-configuration search space is declared, and a correctly specified search of 1,312 configurations built from published specifications finds nothing exceeding what the same search attains on block-bootstrapped noise. An undeclared search space causes the correction to be applied to the wrong denominator, and no amount of care in applying the correction itself fixes that.

Both failures are cheap to detect and both would have been caught by the same discipline: inspect the inputs before trusting the diagnostics, and declare the complete search space before correcting for it. We report both because a paper that only demonstrated the successful correction of a multiple-testing problem would have missed the two failures that mattered most in our own analysis.

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Abbreviations

The following abbreviations are used in this manuscript:

ANOVA	Analysis of variance
CAGR	Compound annual growth rate
CC	Close-to-close
CME	Chicago Mercantile Exchange
CO	Close-to-open
DSR	Deflated Sharpe Ratio
DXY	U.S. Dollar Index
FRED	Federal Reserve Economic Data
OC	Open-to-close
PBO	Probability of backtest overfitting
PSR	Probabilistic Sharpe Ratio
RC	Reality Check (White's Reality Check)
SPA	Superior predictive ability (Hansen's test)
SR	Sharpe ratio

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